Chapter 7. Finite Fourier Analysis

Fourier analysis on Z(N) and finite abelian groups

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7.1 Fourier analysis on $\mathbb{Z}(N)$

Motivation. Replacing the circle by the N-th roots of unity, Fourier series become finite sums: inversion and Parseval are plain linear algebra with an orthogonal basis of exponentials. Splitting into even and odd indices computes all N coefficients in $O(N \log N)$ operations (the fast Fourier transform).

7.1.1 The group $\mathbb{Z}(N)$

For $N \geq 1$ let $\zeta = e^{2\pi i/N}$. The N-th roots of unity $\{1, \zeta, \ldots, \zeta^{N-1}\}$ form a group under multiplication, and $\zeta^k \mapsto k$ identifies it with $\mathbb{Z}(N) = \mathbb{Z}/N\mathbb{Z}$ under addition mod N. Functions on $\mathbb{Z}(N)$ are N-periodic functions on $\mathbb{Z}$; one can view $\mathbb{Z}(N)$ as N equally spaced points on the circle (*discretization*).

7.1.2 Fourier inversion theorem and Plancherel identity on $\mathbb{Z}(N)$

V, the complex functions on $\mathbb{Z}(N)$, is an N-dimensional vector space with inner product $(F, G) = \sum_{k=0}^{N-1} F(k)\overline{G(k)}$ and norm $\|F\|^2 = \sum |F(k)|^2$. For $\ell = 0, \ldots, N-1$ let $e_\ell(k) = \zeta^{\ell k}$.

Lemma 1.1

The family $\{e_0, \ldots, e_{N-1}\}$ is orthogonal: $(e_m, e_\ell) = N$ if $m = \ell$ and 0 if $m \neq \ell$. Hence $\{e_\ell/\sqrt{N}\}$ is an orthonormal basis of V.

Proof. $(e_m, e_\ell) = \sum_{k=0}^{N-1} \zeta^{(m-\ell)k}$. If $m = \ell$ this is N. Otherwise $q = \zeta^{m-\ell} \neq 1$ and $q^N = 1$, so the geometric sum is $\frac{1-q^N}{1-q} = 0$. N orthogonal nonzero vectors in an N-dimensional space form a basis. ■

The n-th **Fourier coefficient** of F is $a_n = \frac{1}{N}(F, e_n) = \frac{1}{N}\sum_{k=0}^{N-1} F(k)\zeta^{-kn}$.

Theorem 1.2

If F is a function on $\mathbb{Z}(N)$, then $F(k) = \sum_{n=0}^{N-1} a_n \zeta^{nk}$ (inversion) and $\sum_{n=0}^{N-1} |a_n|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |F(k)|^2$ (Parseval).

Proof. Expand F in the orthonormal basis $e_n^* = e_n/\sqrt{N}$: $F = \sum (F, e_n^*)e_n^* = \sum \frac{1}{N}(F, e_n)e_n = \sum a_n e_n$. Pythagoras gives $\|F\|^2 = \sum |(F, e_n^*)|^2 = \sum N|a_n|^2$. ■

(Discrete analogue of Fourier series: if $F(k) = f(k/N)$ for a 1-periodic f, then a_n is a Riemann sum for $\hat{f}(n)$.)

7.1.3 The fast Fourier transform

Computing all a_n naively takes about N^2 operations. Let $\#(M)$ be the minimal number of operations (additions and multiplications) needed to compute all Fourier coefficients of any function on $\mathbb{Z}(M)$, here with $\omega_M = e^{-2\pi i/M}$, so that $a_k^M(F) = \frac{1}{M}\sum_r F(r)\omega_M^{kr}$.

Theorem 1.3

For $N = 2^n$, the Fourier coefficients of a function on $\mathbb{Z}(N)$ can be computed with at most $4 \cdot 2^n n = 4N \log_2 N = O(N \log N)$ operations, given ω_N.

Lemma 1.4

Given ω_{2M}, $\#(2M) \le 2\#(M) + 8M$.

Proof. Computing $\omega_{2M}, \omega_{2M}^2, \dots, \omega_{2M}^{2M}$ takes $\le 2M$ operations, and gives $\omega_M = \omega_{2M}^2$. For F on $\mathbb{Z}(2M)$ put $F_0(r) = F(2r)$, $F_1(r) = F(2r+1)$, functions on $\mathbb{Z}(M)$. Splitting the sum into even and odd indices,

$$a_k^{2M}(F) = \frac{1}{2M}\sum_{r=0}^{2M-1} F(r)\omega_{2M}^{kr} = \frac{1}{2}\Big(\frac{1}{M}\sum_{\ell} F_0(\ell)\omega_M^{k\ell} + \frac{1}{M}\sum_{m} F_1(m)\omega_M^{km}\omega_{2M}^{k}\Big)$$
$$= \frac{1}{2}\big(a_k^M(F_0) + a_k^M(F_1)\omega_{2M}^k\big)$$

(with a_k^M read M-periodically in k). Computing all $a^M(F_0)$, $a^M(F_1)$ costs $2\#(M)$; then each of the $2M$ coefficients costs 3 operations. So $\#(2M) \le 2M + 2\#(M) + 6M$. ∎

Proof of Theorem 1.3. Induction on n. For $N = 2$: $a_0 = \frac{1}{2}(F(0)+F(1))$, $a_1 = \frac{1}{2}(F(0)-F(1))$, at most $5 \le 8$ operations. If $\#(2^{n-1}) \le 4\cdot 2^{n-1}(n-1)$, then by Lemma 1.4, $\#(2^n) \le 2\cdot 4\cdot 2^{n-1}(n-1) + 8\cdot 2^{n-1} = 4\cdot 2^n n$. ∎

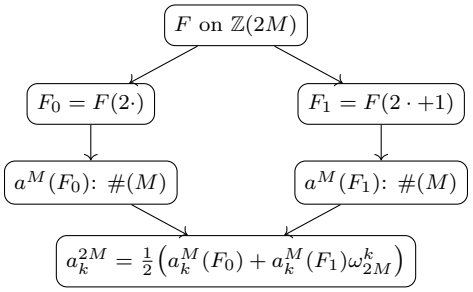


Figure. *One step of the fast Fourier transform (Lemma 1.4).*

7.2 Fourier analysis on finite abelian groups

Motivation. On a general finite abelian group the role of the exponentials is played by characters, homomorphisms into the unit circle. They are orthonormal, and there are enough of them to form a basis (simultaneous diagonalization of the translation operators). This gives Fourier inversion and Plancherel on every finite abelian group, the tool needed for Dirichlet's theorem.

7.2.1 Abelian groups

An **abelian group** $(G, \cdot)$ is associative and commutative with identity 1 and inverses. Examples: $\mathbb{Z}(N)$; the circle $\mathbb{R}/\mathbb{Z}$; $\mathbb{R}$, $\mathbb{R}^+ = (0,\infty)$ under multiplication; $\mathbb{Z}^*(q)$, the units of $\mathbb{Z}(q)$, i.e. residues coprime to q, under multiplication. A **homomorphism** $f : G \to H$ satisfies $f(ab) = f(a)f(b)$; an **isomorphism** is a bijective homomorphism. The **direct product** $G_1 \times G_2$ has componentwise multiplication. (E.g. $\mathbb{Z}^*(8) \cong \mathbb{Z}(2)\times\mathbb{Z}(2)$, while $\mathbb{Z}^*(5) \cong \mathbb{Z}(4)$.) Every finite abelian group is a direct product of cyclic groups; we will not need this.

7.2.2 Characters

A **character** of G is a function $e : G \to S^1 = \{|z| = 1\}$ with $e(ab) = e(a)e(b)$, i.e. a homomorphism $G \to S^1$. The **trivial character** is $e \equiv 1$. $\hat{G}$ denotes the set of characters. On $\mathbb{Z}(N)$ the characters are exactly $e_\ell(k) = \zeta^{\ell k}$ (a character is determined by $e(1)$, an N-th root of unity).

Lemma 2.1

$\hat{G}$ is an abelian group under $(e_1e_2)(a) = e_1(a)e_2(a)$.

Proof. Products of characters are characters; the trivial character is the identity; the inverse of e is $\bar{e} = 1/e$ (as $|e| = 1$); associativity and commutativity are pointwise. ■

Lemma 2.2

If G is finite and $e : G \to \mathbb{C} - \{0\}$ satisfies $e(ab) = e(a)e(b)$, then e is a character.

Proof. G is finite, so $|e(a)|$ is bounded above and below by positive numbers. If $|e(a)| > 1$, then $|e(a^n)| = |e(a)|^n \to \infty$, impossible; if $|e(a)| < 1$, $|e(a^n)| \to 0$, also impossible. So $|e(a)| = 1$. ■

7.2.3 The orthogonality relations

On the space V of complex functions on finite G use $(f, g) = \frac{1}{|G|}\sum_{a\in G} f(a)\overline{g(a)}$.

Theorem 2.3

The characters of G form an orthonormal family in V.

Lemma 2.4

If e is a nontrivial character, then $\sum_{a\in G} e(a) = 0$.

Proof. Choose b with $e(b) \neq 1$. Since $a \mapsto ba$ permutes G, $e(b)\sum_a e(a) = \sum_a e(ba) = \sum_a e(a)$; so $\sum_a e(a) = 0$. ■

Proof of Theorem 2.3. $(e, e) = \frac{1}{|G|}\sum |e(a)|^2 = 1$. For $e \neq e'$, $e\overline{e'} = e(e')^{-1}$ is a nontrivial character, so $(e, e') = \frac{1}{|G|}\sum_a e(a)\overline{e'(a)} = 0$ by Lemma 2.4. ■

Consequently $|\hat{G}| \leq |G|$ (orthonormal vectors are independent).

7.2.4 Characters as a total family

Theorem 2.5

The characters of a finite abelian group G form a basis of the vector space V of functions on G; in particular $|\hat{G}| = |G|$.

Lemma 2.6

A commuting family $\{T_1, \ldots, T_k\}$ of unitary transformations of a finite-dimensional inner product space V ($T_iT_j = T_jT_i$) is simultaneously diagonalizable: V has a basis of common eigenvectors.

Proof. Induction on k. For $k = 1$ this is the spectral theorem for unitary operators. Write $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_s}$, the eigenspaces of T_k. Each T_j, $j < k$, maps V_{λ_i} into itself: if $T_kv = \lambda_i v$, then $T_kT_jv = T_jT_kv = \lambda_iT_jv$. The restrictions of $T_1, \ldots, T_{k-1}$ to V_{λ_i} are commuting unitary operators, so by induction V_{λ_i} has a basis of common eigenvectors. Their union is the desired basis of V. ■

Proof of Theorem 2.5. For $a \in G$ define T_a on V by $(T_a f)(x) = f(ax)$. $T_a T_b = T_b T_a$ (G abelian), and T_a is unitary since $x \mapsto ax$ permutes G. By Lemma 2.6 there is a basis $\{v_b\}$ of common eigenvectors, $T_a v = \lambda_a v$. For such v: $v(1) \neq 0$, since otherwise $v(a) = (T_a v)(1) = \lambda_a v(1) = 0$ for all a. Let $w(x) = v(x)/v(1)$. Then $w(x) = \lambda_x$ (as $v(x) = (T_x v)(1) = \lambda_x v(1)$), so $w(ab) = \lambda_{ab}$, and $\lambda_{ab} v = T_{ab} v = T_a T_b v = \lambda_a \lambda_b v$ gives $w(ab) = w(a)w(b)$; also $w(x) \neq 0$ (as T_x is invertible). By Lemma 2.2, w is a character. The basis $\{v_b\}$ consists of multiples of characters, so the characters span V; with Theorem 2.3 they form a basis. ∎

7.2.5 Fourier inversion and Plancherel formula

For f on G and $e \in \hat{G}$, the **Fourier coefficient** is $\hat{f}(e) = (f, e) = \frac{1}{|G|} \sum_{a \in G} f(a)\overline{e(a)}$, and the **Fourier series** is $f \sim \sum_{e \in \hat{G}} \hat{f}(e) e$.

Theorem 2.7

The characters of a finite abelian group G form an orthonormal basis of V with $(f, g) = \frac{1}{|G|} \sum_a f(a)\overline{g(a)}$. For every f on G, $f = \sum_{e \in \hat{G}} \hat{f}(e) e$ (inversion) and $\|f\|^2 = \sum_{e \in \hat{G}} |\hat{f}(e)|^2$ (Plancherel).

Proof. Theorems 2.3 and 2.5; then expand in the orthonormal basis and apply Pythagoras. ∎

For $G = \mathbb{Z}(N)$ this recovers Theorem 1.2 (up to the normalization of the inner product).

7.3 Selected exercises

Motivation. Three exercises from the book linking the finite theory to the circle (decay of discrete coefficients, sampling of trigonometric polynomials) and recasting finite Fourier series as convolution with a "delta" kernel.

Exercise 7.2 (Decay of discrete Fourier coefficients)

Let f be a C^1 function on the circle and, for $N \geq 1$, $a_N(n) = \frac{1}{N} \sum_{k=1}^{N} f(e^{2\pi i k/N}) e^{-2\pi i k n/N}$. Prove that there is c, independent of N and n, with $|a_N(n)| \leq c/|n|$ whenever $0 < |n| \leq N/2$.

Solution. Write $g(x) = f(e^{2\pi i x})$, a 1-periodic C^1 function, and $M = \sup |g'|$.

Step 1. For an integer ℓ, reindexing $k \mapsto k - \ell$ and using that $k \mapsto g(k/N) e^{-2\pi i k n/N}$ is N-periodic in k,

$$\frac{1}{N} \sum_{k=1}^{N} g\Big(\frac{k+\ell}{N}\Big) e^{-2\pi i k n/N} = e^{2\pi i \ell n/N} a_N(n).$$

Subtracting this from the definition of $a_N(n)$,

$$a_N(n)\big(1 - e^{2\pi i \ell n/N}\big) = \frac{1}{N} \sum_{k=1}^{N} \Big[g\Big(\frac{k}{N}\Big) - g\Big(\frac{k+\ell}{N}\Big) \Big] e^{-2\pi i k n/N}.$$

Step 2. By the mean value theorem each bracket is at most $M\ell/N$ in modulus, so the right side is at most $M\ell/N$.

Step 3. Choose $\ell = \lfloor N/(2|n|) \rfloor$, which is ≥ 1 since $|n| \leq N/2$. Then $\ell|n|/N \leq \frac{1}{2}$, and also $\ell|n|/N \geq |n|/N$ and $\ell|n|/N > \frac{1}{2} - |n|/N$; the larger of these two lower bounds is at least $\frac{1}{4}$. So $s = \ell n/N$ has $\frac{1}{4} \leq |s| \leq \frac{1}{2}$ and

$$|1 - e^{2\pi i s}| = 2|\sin(\pi s)| \geq 2\sin(\pi/4) = \sqrt{2}.$$

Step 4. Combining, $|a_N(n)| \le \frac{M\ell}{\sqrt{2}N} \le \frac{M}{2\sqrt{2}\,|n|}$, using $\ell \le N/(2|n|)$. Take $c = M/(2\sqrt{2})$. ∎

Exercise 7.8 (Sampling a trigonometric polynomial)

Let $P(x) = \sum_{n=1}^{N} a_n e^{2\pi i n x}$.
(a) Show that $\int_0^1 |P(x)|^2 dx = \frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2$.
(b) Prove that $P(x) = \sum_{j=1}^{N} P(j/N) K(x - j/N)$, where

$$K(x) = \frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n x} = \frac{e^{2\pi i x}}{N}\,\frac{1 - e^{2\pi i N x}}{1 - e^{2\pi i x}} \quad (x \notin \mathbb{Z}),$$

and that $K(0) = 1$, $K(j/N) = 0$ for $j \not\equiv 0 \pmod N$.

Solution. Let $\zeta = e^{2\pi i/N}$, so $P(j/N) = \sum_{n=1}^{N} a_n \zeta^{nj}$.

Step 1. For integers $n, m \in \{1, \dots, N\}$, $\frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \mathbb{1}\{n = m\}$: the sum over $j = 1, \dots, N$ equals the sum over $j = 0, \dots, N-1$ ($\zeta^N = \zeta^0$), and $n - m \in (-N, N)$ is $\equiv 0 \pmod N$ only if $n = m$; apply Lemma 1.1.

Step 2. Hence

$$\frac{1}{N}\sum_{j=1}^{N} P(j/N)\zeta^{-mj} = \sum_{n=1}^{N} a_n \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = a_m, \qquad 1 \le m \le N.$$

(a) On the circle, $\int_0^1 e^{2\pi i(n-m)x} dx = \mathbb{1}\{n = m\}$, so $\int_0^1 |P|^2 = \sum_{n,m} a_n \bar{a}_m \int_0^1 e^{2\pi i(n-m)x} dx = \sum_n |a_n|^2$. On $\mathbb{Z}(N)$, by Step 1,

$$\frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2 = \sum_{n,m} a_n \bar{a}_m \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \sum_{n} |a_n|^2.$$

(These are the Parseval identities on the circle and, by Step 2, on $\mathbb{Z}(N)$, Theorem 1.2.) The two sides agree.

(b) Insert Step 2 into the definition of P:

$$P(x) = \sum_{n=1}^{N} \Big(\frac{1}{N}\sum_{j=1}^{N} P(j/N) e^{-2\pi i n j/N}\Big) e^{2\pi i n x} = \sum_{j=1}^{N} P(j/N)\,\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n (x - j/N)},$$

which is $\sum_j P(j/N) K(x - j/N)$. For $x \notin \mathbb{Z}$, $q = e^{2\pi i x} \ne 1$ and $\sum_{n=1}^{N} q^n = q\frac{1-q^N}{1-q}$, giving the closed form. Finally $K(0) = \frac{1}{N}\cdot N = 1$, and for $j \not\equiv 0$ the number $q = \zeta^j$ satisfies $q \ne 1$, $q^N = 1$, so $K(j/N) = \frac{1}{N}\sum_{n=1}^{N} q^n = \frac{q}{N}\frac{1-q^N}{1-q} = 0$. Thus P is determined by its N samples $P(j/N)$. ∎

Exercise 7.13 (Convolution and the delta kernel on G)

Let G be a finite abelian group with unit 1_G and $(f * g)(a) = \frac{1}{|G|}\sum_{b \in G} f(b) g(ab^{-1})$.
(a) Show that $\widehat{f * g}(e) = \hat{f}(e)\hat{g}(e)$ for all $e \in \hat{G}$.
(b) Show that $\sum_{e \in \hat{G}} e(c) = 0$ whenever $c \ne 1_G$.
(c) Let $D(c) = \sum_{e \in \hat{G}} e(c)$. Show that the Fourier series $Sf = \sum_e \hat{f}(e) e$ equals $f * D$, and deduce $Sf = f$.

Solution.

(a) Since $\overline{e(bc)} = \overline{e(b)}\,\overline{e(c)}$ and $c = ab^{-1}$ runs over G as a does (for fixed b),

$$\begin{aligned}\widehat{f * g}(e) &= \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{a\in G} g(ab^{-1})\overline{e(a)} = \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{c\in G} g(c)\overline{e(bc)} \\ &= \Big(\frac{1}{|G|}\sum_{b} f(b)\overline{e(b)}\Big)\Big(\frac{1}{|G|}\sum_{c} g(c)\overline{e(c)}\Big) = \hat{f}(e)\hat{g}(e).\end{aligned}$$

(b)

Step 1. There is $e_0 \in \hat{G}$ with $e_0(c) \neq 1$. Otherwise every character takes the same value at c and at 1_G, hence so does every linear combination of characters; by Theorem 2.5 every function is such a combination, but the indicator of $\{1_G\}$ takes the values 1 and 0 there.

Step 2. By Lemma 2.1, $e \mapsto e_0 e$ is a bijection of $\hat{G}$. Hence $\sum_e e(c) = \sum_e (e_0 e)(c) = e_0(c)\sum_e e(c)$, and $e_0(c) \neq 1$ forces $\sum_e e(c) = 0$.

(c)

Step 1. By (b) and $|\hat{G}| = |G|$ (Theorem 2.5), $D(1_G) = |G|$ and $D(c) = 0$ for $c \neq 1_G$.

Step 2. Using $\overline{e(b)} = e(b)^{-1} = e(b^{-1})$,

$$Sf(a) = \sum_{e\in\hat{G}} \frac{1}{|G|}\sum_{b\in G} f(b)\overline{e(b)}e(a) = \frac{1}{|G|}\sum_{b\in G} f(b)\sum_{e\in\hat{G}} e(ab^{-1}) = (f * D)(a).$$

Step 3. By Step 1 only $b = a$ contributes, so $(f * D)(a) = \frac{1}{|G|} f(a)|G| = f(a)$. Hence $Sf = f$, a second proof of the inversion formula in Theorem 2.7. ∎